

Bounded Martin's Maximum, Axiom $(*)$, and Generic $\Pi_2^{H(\omega_2)}$ -maximality

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Abstract

We present a proof that, under large cardinal assumptions, a form of Bounded Martin's Maximum, $\text{UB} - \text{BMM}^{++}$, a form of the axiom $(*)$, $(*)_{\text{UB}}$, and a form of generic $\Pi_2^{H(\omega_2)}$ -maximality, $(\dagger)_{\text{UB}}$, are all equivalent.

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1 Introduction

We wish to present a proof of equivalence of the following three principles (in the context of large cardinals).

Definition 1.1. The sentence $\text{UB} - \text{BMM}^{++}$ asserts that for every universally Baire set A , every Σ_1 formula $\phi(v_a)$ in the language $\{R_{\in}^2, R_{\text{NS}_{\omega_1}}^1, R_A^1\}$ and every $a \in H_{\omega_2}$, if

$$\exists \mathbb{P} \text{ ssp}, V^{\mathbb{P}} \models ((H(\omega_2), \in, \text{NS}_{\omega_1}, A_{\dot{g}}) \models \phi[a]),$$

then $(H(\omega_2), \in, \text{NS}_{\omega_1}, A) \models \phi[a]$. $\square$

Definition 1.2. The sentence $(*)_{\text{UB}}$ asserts that for every universally Baire dense subset D of $\mathbb{P}_{\max}$ and for every $a \subseteq \omega_1$ satisfying that $\omega_1^{L[a]} = \omega_1$, there exist $p \in D$ and a generic iteration $\mathcal{I}$ of p such that $i^{\mathcal{I}}(a^p) = a$ and $i^{\mathcal{I}}(I^p) = \text{NS}_{\omega_1} \cap \mathcal{P}_{\infty}^{\mathcal{I}}$. $\square$

Definition 1.3. The sentence $(\dagger)_{\text{UB}}$ asserts that for every universally Baire set A and every Π_2 sentence ϕ in the language $\{R_{\in}^2, R_{\text{NS}_{\omega_1}}^1, R_A^1\}$, if

$$\exists \mathbb{P}, \Vdash_{\mathbb{P}} ((H(\omega_2), \in, \text{NS}_{\omega_1}, A_{\dot{g}}) \models \phi),$$

then $(H(\omega_2), \in, \text{NS}_{\omega_1}, A) \models \phi$. $\square$

We will assume that there is a proper class of Woodin cardinals. That $\text{UB} - \text{BMM}^{++}$ implies $(*)_{\text{UB}}$ was established in [AS21]. The converse follows by the proof of [Woo10, Theorem 10.127]. We learned a proof that $(*)_{\text{UB}}$ implies $(\dagger)_{\text{UB}}$ from [Lar10], while the converse, in the presence of a supercompact cardinal, was observed in [Kas25].

Theorem 1.4. *Suppose there exists a proper class of Woodin cardinals. Then the following holds.*

- a. $\text{UB} - \text{BMM}^{++}$ and $(*)_{\text{UB}}$ are equivalent.
- b. $(*)_{\text{UB}}$ implies $(\dagger)_{\text{UB}}$.
- c. If $(\dagger)_{\text{UB}}$ holds and if there is a supercompact cardinal, then $(*)_{\text{UB}}$ holds.

Proof. That $\text{UB} - \text{BMM}^{++}$ implies $(*)_{\text{UB}}$ follows from [AS21, Lemma 2.14]. The converse is proved in Theorem 6.5. We prove in Corollary 5.3 that $(*)_{\text{UB}}$ implies $(\dagger)_{\text{UB}}$. Finally, $\mathfrak{c}$ is verified in [Kas25, Proposition 6.2.11]. $\square$

2 Some Lemmas

Lemma 2.1. *Suppose δ is a Woodin cardinal. Then there exists a semi-proper poset that forces that NS_{ω_1} is saturated.*

Proof. See [KV23, Proposition [3.3.6]]. $\square$

Lemma 2.2. *Let λ be the smallest beth-fixed point. Then there exists a c.c.c. poset of size in V_λ that forces MA_{ω_1} .*

Proof. This follows from the proof of [Kun80, Theorem 6.3]. To ensure that there exists a regular $\kappa \geq \omega_1$ satisfying $2^{<\kappa} = \kappa$, force first with $\text{Col}(\omega_2, 2^{\omega_1})$. $\square$

Lemma 2.3. *Suppose*

1. $\mathbb{P}$ is a c.c.c. poset,
2. g is V -generic for $\mathbb{P}$,
3. C is a club subset of ω_1 in $V[g]$.

Then there exists $\bar{C} \in V$ which is a subset of C and which is a club in ω_1 .

Proof. See [Jec86, Part III, Theorem 1.5]. $\square$

Lemma 2.4. *Suppose*

1. I is a normal precipitous ideal on ω_1 ,
2. $\mathbb{P}$ is a c.c.c. poset.

Then it holds in $V^\mathbb{P}$ that the ideal on ω_1 generated by I is precipitous.

Proof. See [Kak81, Theorem 1]. $\square$

Lemma 2.5. *Suppose*

1. NS_{ω_1} is precipitous,
2. $\mathbb{P}$ is a c.c.c. poset.

Then NS_{ω_1} is precipitous in $V^\mathbb{P}$.

Proof. By Lemma 2.4, the ideal generated by $\text{NS}_{\omega_1}^V$ is precipitous in $V^\mathbb{P}$. By Lemma 2.3, this ideal is exactly the non-stationary ideal of the extension. $\square$

Lemma 2.6. *Suppose*

1. δ is a beth-fixed point,
2. A is a $<\delta$ -universally Baire set of reals,
3. $\theta > \delta$ is inaccessible,
4. M is transitive,

5. $\pi : (M, \in, \bar{\delta}, \bar{A}) \longrightarrow (H_\theta, \in, \delta, A)$ is elementary,

6. g is $<\bar{\delta}$ -generic over M .

Then $\bar{A}_g = A \cap M[g]$.

Proof.

1° Let $(T, S) \in \text{ran}(\pi)$ be a pair of $<\delta$ -absolutely complementing trees satisfying that $p[T] = A$ and let

$$(\bar{T}, \bar{S}) := \pi^{-1}((T, S)).$$

Note that $p[\bar{T}] \subseteq p[T]$ and $p[\bar{S}] \subseteq p[S]$.

2° Since $p[\bar{T}]^{M[g]} = \mathbb{R}^{M[g]} - p[\bar{S}]^{M[g]}$, it follows that

$$p[\bar{T}]^{M[g]} = p[T] \cap M[g] \text{ and } p[\bar{S}]^{M[g]} = p[S].$$

3° Thus, $\bar{A}_g = p[\bar{T}]^{M[g]} = p[T] \cap M[g] = A \cap M[g]$.

□

Lemma 2.7. *Suppose*

1. δ is a limit of Woodin cardinals,
2. A is a $<\delta$ -universally Baire set of reals,
3. $\phi(x)$ is a formula in the language $\{R_\infty^2, R_A^1\}$,
4. $B := \{b \in \mathbb{R} : (H_{\omega_1}, \in, A) \models \phi[b]\}$.

Then the set B universally Baire and for every poset $\mathbb{P} \in V_\delta$,

$$V^\mathbb{P} \models B_{\dot{g}} = \{b \in \mathbb{R} : (H_{\omega_1}, \in, A_{\dot{g}}) \models \phi[b]\}.$$

Proof. See [Kas22a, Theorem 1.4-XI].

□

Proposition 2.8. *Suppose there is a proper class of Woodin cardinals. Then for every universally Baire set A ,*

$$\mathcal{P}(\mathbb{R}) \cap L(\mathbb{R}, A) \subseteq \text{UB}.$$

□

3 Pre-conditions

Lemma 3.1. *Suppose I is a normal precipitous ideal on ω_1 . Then (V, I) is generically Ord-iterable in every generic extension.*

Proof. See [Kas22b, Proposition 2-III]. □

Corollary 3.2. *Suppose*

- a. M is a transitive model of ZFC,
- b. I is a normal precipitous ideal on ω_1 in M .

Then (M, I) is generically $o(M)$ -iterable. □

Corollary 3.3. *Suppose*

- a. M and $\bar{M}$ are transitive models of ZFC,
- b. I is a normal precipitous ideal on ω_1 in M ,
- c. γ is an inaccessible cardinal in $\bar{M}$,
- d. $\pi : \bar{M} \rightarrow M$ is elementary embedding with critical point at least γ .

Then $(\bar{M}, I)$ is generically $o(M)$ -iterable. □

Lemma 3.4. *Suppose*

- 1. κ is measurable,
- 2. $\theta > \kappa$ is inaccessible,
- 3. $M \prec (H_\theta, \in, \kappa)$ is of cardinality $< \kappa$,
- 4. $\alpha < \kappa$.

Then there exists $N \prec H_\theta$ such that

- a. $|N| < \kappa$,
- b. $M \subseteq N$,
- c. $\kappa \cap M$ is an initial segment of $\kappa \cap N$,
- d. $\text{otp}(\kappa \cap N) > \alpha$.

Proof.

1° It suffices to show that there exists $N \prec H_\theta$ such that

- a. $|N| < \kappa$,
- b. $M \subseteq N$,
- c. $\kappa \cap M$ is an initial segment of $\kappa \cap N$,

d. $\text{otp}(\kappa \cap N) > \text{otp}(\kappa \cap M)$.

2° Let $\mu \in M$ be a normal measure on κ , let $\gamma := \min \bigcap (\mu \cap M)$, and let $N := \text{Hull}(M, \{\gamma\})$. We will be done if we show that $\kappa \cap M$ is an initial segment of $\kappa \cap N$.

3° Let $\xi \in \kappa \cap N$ and $\eta \in \kappa \cap M$ be such that $\xi < \eta$. We want to show that $\xi \in M$.

4° There exists $f : \kappa \rightarrow H_\theta$ such that $f \in M$ and $f(\gamma) = \xi$.

5° We may assume without loss of generality that $f : \kappa \rightarrow \eta$.

6° Since $\eta < \kappa$, there exists a unique $\xi_0 < \eta$ such that

$$\{\zeta < \kappa : f(\zeta) = \xi_0\} \in \mu.$$

7° Note that $\xi_0 \in M$, so

$$\{\zeta < \kappa : f(\zeta) = \xi_0\} \in \mu \cap M.$$

8° This means that

$$\gamma \in \{\zeta < \kappa : f(\zeta) = \xi_0\},$$

so $\xi = f(\gamma) = \xi_0 \in M$, as required.

□

Proposition 3.5. *Suppose*

1. κ is measurable,
2. $\theta > \kappa$ is inaccessible,
3. M is a transitive set of cardinality $< \kappa$,
4. $\pi : (M, \in, \bar{\kappa}) \rightarrow (H_\theta, \in, \kappa)$ is elementary,
5. g is a $< \bar{\kappa}$ -generic over M ,
6. I is a normal precipitous ideal on ω_1 in $M[g]$.

Then $(M[g], I)$ is generically iterable.

Proof.

1° By Lemma 3.4, there exists $X \prec H_\theta$ such that

- a. $|X| < \kappa$,
- b. $\text{ran}(\pi) \subseteq X$,
- c. $\kappa \cap \text{ran}(\pi)$ is an initial segment of $\kappa \cap X$,
- d. $\text{otp}(\kappa \cap X) > \omega_1$.

- 2° Let N be the transitive collapse of X and let $\pi : M \longrightarrow N$ be the elementary embedding induced by the inclusion $\text{ran}(M) \prec X$. Note that $\bar{\kappa}$ is inaccessible in M and $\text{crit}(\pi) \geq \bar{\kappa}$.
- 3° Let $\pi[g] : M[g] \longrightarrow N[g]$ be the elementary embedding induced by π . Corollary 3.3 now implies that $(M[g], I)$ is generically $o(N[g])$ -iterable. In particular, it is generically ω_1 -iterable, as required.

□

Definition 3.6. A pair (M, I) is a **pre-condition** iff

- a. M is a countable transitive model of ZFC,
- b. I is a normal precipitous ideal on ω_1 in M ,
- c. (M, I) is generically iterable.

□

Lemma 3.7. *Suppose*

1. ZFC holds,
2. J is a normal ideal on ω_1 ,
3. (M, I) is a pre-condition.

Then there exists a generic iteration $i : (M, I) \longrightarrow (M^, I^*)$ of length ω_1 such that $I^* = J \cap M^*$.*

Proof. See [Kas22b, Proposition 2-X].

□

Definition 3.8. Suppose

1. A is a set of reals,
2. (M, I) is a pre-condition.

Then (M, I) is **A-iterable** iff

- a. $A \cap M \in M$,
- b. for every generic iteration $i : (M, I) \longrightarrow (N, J)$, $i(A \cap M) = A \cap N$.

□

Lemma 3.9. *Suppose*

1. δ is a limit of Woodin cardinals,
2. $\gamma < \delta$ is inaccessible,
3. $T, S \in V_\delta$ are γ^+ -homogeneous trees,
4. $p[T] = \mathbb{R} - p[S]$,
5. $\theta > \delta$ is inaccessible,

6. $M \prec (H_\theta, \in, \gamma, \delta, T, S)$ is of cardinality at most γ .

Then there exists $X \prec H_\theta$ such that

- a. $|X| \leq 2^\omega |M|$,
- b. $M \subseteq X$,
- c. $X \cap V_\gamma = M \cap V_\gamma$,
- d. letting $(\bar{\cdot}) : X \cup \{X\} \longrightarrow \bar{X} \cup \{\bar{X}\}$ be the transitive collapse,

$$p[\bar{T}] = p[T] \text{ and } p[\bar{S}] = p[S].$$

Proof.

1° It suffices to show that for a given $x \in \mathbb{R}$, there exists $X \prec H_\theta$ such that

- a. $|X| = |M|$,
- b. $M \subseteq X$,
- c. $X \cap V_\gamma = M \cap V_\gamma$,
- d. if $x \in p[T]$, then $x \in p[\bar{T}]$,
- e. if $x \in p[S]$, then $x \in p[\bar{S}]$.

2° Let $\mu, \nu \in M$ be γ^+ -homogeneity systems for T, S , respectively. Let ξ and U be defined as follows:

- if $x \in p[T]$, then $\xi := (\mu_{x \upharpoonright n} : n < \omega)$ and $U := T$;
- if $x \in p[S]$, then $\xi := (\nu_{x \upharpoonright n} : n < \omega)$ and $U := S$.

3° For $n < \omega$, let $A_n := \bigcap (\xi_n \cap M) \in \xi_n$. Since $x \in p[U]$, there exists an ω -sequence z such that for all $n < \omega$, $z \upharpoonright n \in A_n$.

4° Let $X := \text{Hull}(M, \text{ran}(z)) \prec H_\theta$. It is clear that $|X| = |M|$ and $M \prec X$.

5° **Claim.** $X \cap V_\gamma = M \cap V_\gamma$

Proof.

- 1' Let $w \in X \cap V_\gamma$ and let us show that $w \in M$.
- 2' Let Z be such that for all $n < \omega$, $\xi_n \subseteq \mathcal{P}(Z^n)$. Note that $Z \in M$.
- 3' There exist $n < \omega$ and $f : Z^n \longrightarrow H_\theta$ such that $w = f(z \upharpoonright n)$.
- 4' We may assume without loss of generality that $f : Z^n \longrightarrow V_\gamma$.
- 5' Since ξ_n is γ^+ -complete, there exists a unique $w' \in V_\gamma$ such that

$$\{u \in Z^n : f(u) = w'\} \in \xi_n.$$

6' Note that $w' \in M$ and consequently,

$$\{u \in Z^n : f(u) = w'\} \in \xi_n \cap M.$$

7' Thus,

$$z \restriction n \in \{u \in Z^n : f(u) = w'\},$$

so $w = f(z \restriction n) = w' \in M$, as required.

□

6° Finally, the fact that $\text{ran}(z) \subseteq X$ immediately yields that $(x, (\bar{z}_n)_{n < \omega}) \in \bar{U}$, which means that

- if $x \in p[T]$, then $x \in p[\bar{T}]$,
- if $x \in p[S]$, then $x \in p[\bar{S}]$.

This concludes the proof.

□

Proposition 3.10. *Suppose*

1. δ is a limit of Woodins,
2. A is $<\delta$ -universally Baire,
3. $\theta > \delta$ is inaccessible,
4. M is transitive and countable,
5. $\pi : (M, \in, \bar{\delta}, \bar{A}) \longrightarrow (H_\theta, \in, \delta, A)$ is elementary,
6. g is $<\bar{\delta}$ -generic over M ,
7. I is a precipitous ideal on ω_1 in $M[g]$.

Then $(M[g], I)$ is an A -iterable pre-condition.

Proof.

- 1° Proposition 3.5 implies that $(M[g], I)$ is a pre-condition. It remains to show that it is A -iterable.
- 2° Let $\bar{\gamma} < \bar{\delta}$ be an inaccessible cardinal in M such that g is $<\bar{\gamma}$ -generic over M and let $\bar{M} := M \restriction \bar{\gamma}$. It suffices to show that $(\bar{M}[g], I)$ is A -iterable.
- 3° Lemma 2.6 implies that $A \cap \bar{M}[g] = \bar{A}_g \in \bar{M}[g]$.
- 4° Let $i : (\bar{M}[g], I) \longrightarrow (\bar{N}, J)$ be a generic iteration of countable length. We will be done if we show that $i(A \cap \bar{M}[g]) = A \cap \bar{N}$.

5° Let $T, S \in \text{ran}(\pi)$ be γ^+ -homogeneous trees satisfying that $p[T] = A$ and $p[S] = \mathbb{R} - A$.

6° By Lemma 3.9, there exists $X \prec H_\theta$ such that, letting

$$\pi' : (M', \in, \bar{T}, \bar{S}) \longrightarrow (X, \in, T, S)$$

be the anti-collapse,

- a. $|M'| \leq 2^\omega$,
- b. $\text{ran}(\pi) \subseteq X$,
- c. $X \cap V_\gamma = \text{ran}(\pi) \cap V_\gamma$,
- d. $p[\bar{T}] = A$ and $p[\bar{S}] = \mathbb{R} - A$.

Note that $\bar{\gamma}$ is inaccessible in M' and $M' \restriction \bar{\gamma} = \bar{M}$.

7° By Proposition 3.5, there exists a generic iteration

$$i' : (M'[g], I) \longrightarrow (N, J)$$

such that the generics used in it are the same as those used in i .¹

8° Observe that $N \restriction i'(\bar{\gamma}) = \bar{N}$ and $i \restriction \bar{M}[g] = i$. This means that we will be done if we show that $i'(A \cap M'[g]) = A \cap N$.

9° Since $p[\bar{T}] \subseteq p[i'(\bar{T})]$ and $p[\bar{S}] \subseteq p[i'(\bar{S})]$ and

$$p[\bar{T}] = \mathbb{R} - p[\bar{S}],$$

we conclude that $p[\bar{T}] = p[i'(\bar{T})]$.

10° Thus,

$$i'(A \cap M'[g]) = i'(p[\bar{T}]^{M'[g]}) = p[i'(\bar{T})]^N = p[\bar{T}] \cap N = A \cap N,$$

as required. □

Lemma 3.11. *Suppose*

1. δ is a limit of Woodins,
2. A is $<\delta$ -universally Baire,
3. $\theta > \delta$ is inaccessible,
4. M is transitive and countable,

¹In other words, when using these generics on $M'[g]$, we keep producing well-founded models.

5. $\pi : (M, \in, \bar{\delta}, \bar{A}) \longrightarrow (H_\theta, \in, \delta, A)$ is elementary,
6. N is a forcing extension of M by a poset in $M \restriction \bar{\delta}$.

Then $(H_{\omega_1}^N, \in, A \cap N) \prec (H_{\omega_1}, \in, A)$.

Proof.

- 1° We apply the Tarski-Vaught test. Let $\phi(v_a, v_b)$ be a formula in the language $\{R_\in^2, R_A^1\}$ and let $a \in H_{\omega_1}^N$. Suppose that

$$(H_{\omega_1}, \in, A) \models (\exists v_b, \phi)[a].$$

We want to find $b \in H_{\omega_1}^N$ such that

$$(H_{\omega_1}, \in, A) \models \phi[a, b].$$

- 2° We may assume without loss of generality that $a \in \mathbb{R}$ and

$$(H_{\omega_1}, \in, A) \models (\exists v_b \in \mathbb{R}, \phi)[a].$$

- 3° Let $C := \{(x, y) \in \mathbb{R} \times \mathbb{R} : (H_{\omega_1}, \in, A) \models \phi[x, y]\}$. We will be done if we show that for some $b \in \mathbb{R} \cap N$, $(a, b) \in C \cap N$.

- 4° Let $E := \{x \in \mathbb{R} : (H_{\omega_1}, \in, A) \models (\exists v_b, \phi)[x]\}$. Observe that $a \in E$ and

$$(H_{\omega_1}, \in, A, C, E) \models \forall x \in R_E^1, \exists y, (x, y) \in R_C^1.$$

- 5° Since $C, E \in M$, there are $\bar{C}, \bar{E}$ such that $\pi : \bar{C}, \bar{E} \mapsto C, E$.

- 6° By Lemma 2.7, the sets C, E are $<\delta$ -universally Baire.

- 7° By elementarity, $\bar{C}, \bar{E}$ are $<\bar{\delta}$ -universally Baire in M and

$$(H_{\omega_1}^M, \in, \bar{A}, \bar{C}, \bar{E}) \models \forall x \in R_{\bar{E}}^1, \exists y, (x, y) \in R_{\bar{C}}^1.$$

- 8° Let g be a $<\bar{\delta}$ -generic over M such that $N = M[g]$. By Lemma 2.7,

$$(H_{\omega_1}^N, \in, \bar{A}_g, \bar{C}_g, \bar{E}_g) \models \forall x \in R_{\bar{E}}^1, \exists y, (x, y) \in R_{\bar{C}}^1.$$

- 9° By Lemma 2.6, $\bar{C}_g = C \cap N$ and $\bar{E}_g = E \cap N$, which means that

$$\forall x \in E \cap N, \exists y \in N, (x, y) \in C \cap N.$$

- 10° Since $a \in E \cap N$, we conclude that

$$\exists y \in N, (a, y) \in C \cap N,$$

as required. □

4 Certification by $\mathbb{P}_{\max}$ -conditions

Notation 4.1. Suppose $p \in \mathbb{P}_{\max}$ and $\mathcal{I}$ is a (generic) iteration of p of length at most $\omega_1 + 1$. Then

- a. for $\alpha < \text{len}(\mathcal{I})$, $\mathcal{P}_\alpha^\mathcal{I}$ denotes the α^{th} structure in the iteration $\mathcal{I}$,
- b. for $\alpha \leq \beta < \text{len}(\mathcal{I})$, $i_{\alpha,\beta}^\mathcal{I}$ denotes the embedding $\mathcal{P}_\alpha^\mathcal{I} \longrightarrow \mathcal{P}_\beta^\mathcal{I}$ given by $\mathcal{I}$,
- c. $\mathcal{P}_\infty^\mathcal{I}$ denotes the direct limit of the iteration $\mathcal{I}$,
- d. for $\alpha < \text{len}(\mathcal{I})$, $i_{\alpha,\infty}^\mathcal{I}$ denotes the direct limit embedding $\mathcal{P}_\alpha^\mathcal{I} \longrightarrow \mathcal{P}_\infty^\mathcal{I}$ given by $\mathcal{I}$,
- e. $i^\mathcal{I}$ denotes $i_{0,\infty}^\mathcal{I}$ □

Definition 4.2. Suppose A is a set of reals and $p \in \mathbb{P}_{\max}$. Then p is *A-iterable* iff (M^p, I^p) is an A -iterable pre-condition. □

Definition 4.3. Suppose A is a set of reals and $p \in \mathbb{P}_{\max}$. Then p *certifies* A iff

- a. p is A -iterable,
- b. $(H_{\omega_1}^p, \in, A \cap p) \prec (H_{\omega_1}, \in, A)$. □

Proposition 4.4. Suppose there is a proper class of Woodin cardinals. Then every universally Baire set of reals is certified by a $\mathbb{P}_{\max}$ -condition.

Proof.

- 1° Let A be an arbitrary universally Baire set. We want to find $p \in \mathbb{P}_{\max}$ that certifies A .
- 2° Let δ be a limit of Woodin cardinals, let $\theta > \delta$ be inaccessible, and let

$$\pi : (M, \in, \bar{A}, \bar{\delta}) \longrightarrow (H_\theta, \in, A, \delta)$$

be elementary and countable such that M is transitive.

- 3° By Lemma 2.1, there exists a $<\bar{\delta}$ -generic g over M such that NS_{ω_1} is saturated in $M[g]$; and by Lemmas 2.2 and 2.5, there exists a $<\bar{\delta}$ -generic h over $M[g]$ such that MA_{ω_1} holds and NS_{ω_1} is precipitous in $M[g][h]$.

- 4° Let a^p be such that $M[g][h] \models (\omega_1^{L[a^p]} = \omega_1)$ and let

$$p := (M[g][h], \in, \text{NS}_{\omega_1}^{M[g][h]}, a^p).$$

By Proposition 3.5, p is a $\mathbb{P}_{\max}$ condition.

- 5° By Proposition 3.10, p is A -iterable.

6° By Lemma 3.11, $(H_{\omega_1}^p, \in, A \cap p) \prec (H_{\omega_1}, \in, A)$. This complete the verification that p certifies A . □

Notation 4.5. Suppose $p \in \mathbb{P}_{\max}$ and G is V -generic for $\mathbb{P}_{\max}$ containing p . Then $\mathcal{I}_{p,G}$ denotes the iteration of p given by G . □

Lemma 4.6. *Suppose*

1. ZFC holds,
2. every real has a dagger,
3. J is a normal ideal on ω_1 ,
4. for all $\alpha < \omega_1$, D_α is a dense subset of $\mathbb{P}_{\max}$,
5. $p \in \mathbb{P}_{\max}$.

Then there exists a function $f : \omega_1 \longrightarrow \mathbb{P}_{\max}$ such that

- a. $f(0) = p$,
- b. for all $\alpha < \omega_1$, $f(\alpha + 1) < f(\alpha)$,
- c. for all $\alpha < \omega_1$, $f(\alpha + 1) \in D_\alpha$,
- d. letting $\mathcal{I}$ be the union of the iterations $(\mathcal{I}_{p,f(\alpha)} : \alpha < \omega_1)$, it holds that $i^{\mathcal{I}}(I^p) = J \cap \mathcal{P}_\infty^{\mathcal{I}}$.

Proof. See [Kas22b, Lemma 4.3-VI]. □

Lemma 4.7. *Suppose*

1. p is a $\mathbb{P}_{\max}$ -condition,
2. $X \in p$ is a set of $\mathbb{P}_{\max}$ -conditions,
3. for all $q \in X$, $q > p$.

Then $p \Vdash i^{\mathcal{I}_{p,\dot{g}}}(P) \subseteq \dot{g}$.

Proof. See [Kas22b, Lemma 4.3-VII]. □

Proposition 4.8. *Suppose*

1. ZF holds,
2. every real has a dagger,
3. every set of reals is certified by a $\mathbb{P}_{\max}$ -condition.

Then it holds in $V^{\mathbb{P}_{\max}}$ that for all $a \in H_{\omega_2}$, there exists $p \in \dot{g}$ such that $a \in \mathcal{P}_\infty^{\mathcal{I}_{p,\dot{g}}}$.

Proof.

1° Let τ be a $\mathbb{P}_{\max}$ name for a subset of ω_1 . We want to show that

$$\Vdash_{\mathbb{P}_{\max}} \exists q \in \dot{g}, \tau \in \mathcal{P}_{\infty}^{\mathcal{I}_{q,\dot{g}}}.$$

2° Let $p \in \mathbb{P}_{\max}$ be arbitrary. We want to find $q \leq p$ and $e \in q$ such that

$$q \Vdash \tau = i^{\mathcal{I}_{q,\dot{g}}}(e).$$

This suffices for the conclusion.

3° Let A be the set

$$\{(q, \alpha, i) : q \in \mathbb{P}_{\max}, q \Vdash \chi_{\tau}(\alpha) = i\},$$

where χ_{τ} denotes the characteristic function of τ in $V_{\max}^{\mathbb{P}}$; and let A^* be the set of reals coding A . Since every set of reals is certified by a $\mathbb{P}_{\max}$ condition, there exists $q \in \mathbb{P}_{\max}$ such that

- a. $p \in q$,
- b. q is A^* -iterable,
- c. $(H_{\omega_2}^q, \in, A^* \cap q) \prec (H_{\omega_2}, \in, A^*)$.

4° For $\alpha < \omega_1$, let

$$D_{\alpha} := \{q \in \mathbb{P}_{\max} : \exists i < 2, (q, \alpha, i) \in A\}.$$

The property [3°c](#) implies that

- a. for all $x \in \mathbb{R} \cap q$, $x^{\dagger} \in q$,
- b. $(D_{\alpha} \cap q : \alpha < \omega_1^q) \in q$,
- c. for all $\alpha < \omega_1^q$, $D_{\alpha} \cap q$ is a dense subset of $\mathbb{P}_{\max} \cap q$.

5° In q , Lemma [4.6](#) implies that there exists a function $f : \omega_1 \longrightarrow \mathbb{P}_{\max}$ such that

- a. $f(0) = p$,
- b. for all $\alpha < \omega_1$, $f(\alpha + 1) < f(\alpha)$,
- c. for all $\alpha < \omega_1$, $f(\alpha + 1) \in D_{\alpha} \cap q$,
- d. letting $\mathcal{I}$ be the union of the iterations $(\mathcal{I}_{p,f(\alpha)} : \alpha < \omega_1)$, it holds that $i^{\mathcal{I}}(I^p) = I^q \cap \mathcal{P}_{\infty}^{\mathcal{I}}$.

6° Let $\mathcal{I}$ be the union of the iterations $(\mathcal{I}_{p,f(\alpha)} : \alpha < \omega_1)$. Up to replacing a^q by $i^{\mathcal{I}}(a^p)$, we may assume that $a^q = i^{\mathcal{I}}(a^p)$. In particular, for all $\alpha < \omega_1^q$, $q < f(\alpha)$.

7° Let $e := \{\alpha < \omega_1^q : (f(\alpha + 1), \alpha, 1) \in D_{\alpha} \cap q\} \in q$. We will be done if we show that $q \Vdash i^{\mathcal{I}_{q,\dot{g}}}(e) = \tau$.

8° **Claim.** Let g be V -generic for $\mathbb{P}_{\max}$ containing q . Then $i^{\mathcal{I}_{q,g}}(e) = \tau^g$.

Proof.

1' By 3°c,

$$q \models \forall \alpha < \omega_1, \forall i < 2,$$

$$\chi_e(\alpha) = i \iff (f(\alpha + 1), \alpha, i) \in A \cap q.$$

2' Let $\hat{f} := i^{\mathcal{I}_{q,g}}(f) : \omega_1 \longrightarrow \mathbb{P}_{\max}$. It follows by elementarity of $i^{\mathcal{I}_{q,g}}$ that

$$\mathcal{P}_{\infty}^{\mathcal{I}_{q,g}} \models \forall \alpha < \omega_1, \forall i < 2,$$

$$\chi_{i^{\mathcal{I}_{q,g}}(e)}(\alpha) = i \iff (\hat{f}(\alpha + 1), \alpha, i) \in i^{\mathcal{I}_{q,g}}(A \cap q).$$

3' Since q is A^* -iterable, it follows that $i^{\mathcal{I}_{q,g}}(A \cap q) = A$. Hence, for all $\alpha < \omega_1$ and all $i < 2$,

$$\chi_{i^{\mathcal{I}_{q,g}}(e)}(\alpha) = i \iff (\hat{f}(\alpha + 1), \alpha, i) \in A.$$

4' This means that for all $\alpha < \omega_1$,

- a. $\alpha \in i^{\mathcal{I}_{q,g}}(e)$ if and only if $\hat{f}(\alpha + 1) \Vdash^V \alpha \in \tau$,
- b. $\alpha \notin i^{\mathcal{I}_{q,g}}(e)$ if and only if $\hat{f}(\alpha + 1) \Vdash^V \alpha \notin \tau$.

5' By Lemma 4.7, $\text{ran}(\hat{f}) \subseteq g$. This and the previous point yield that for all $\alpha < \omega_1$,

$$\alpha \in i^{\mathcal{I}_{q,g}}(e) \iff \alpha \in \tau^g;$$

in other words, $i^{\mathcal{I}_{q,g}}(e) = \tau^g$, as required.

□

9° The claim concludes the proof.

□

5 $(*)_{\text{UB}}$ implies $(\dagger)_{\text{UB}}$

Proposition 5.1. *Suppose*

1. *ZF holds,*
2. *every real has a dagger,*
3. *every set of reals is certified by a $\mathbb{P}_{\text{max}}$ -condition,*
4. *A is a set of reals,*
5. *$\phi(x)$ is a Σ_1 formula in the language $\{R_{\in}^2, R_{\text{NS}_{\omega_1}}^1, R_A^1\}$,*
6. *for all $p \in \mathbb{P}_{\text{max}}$, for all $a \in H_{\omega_2}^p$, there exists an A -iterable $\mathbb{P}_{\text{max}}$ -condition $q \leq p$ such that*

$$(H_{\omega_2}^q, \in, I^q, A \cap q) \models \phi[i^{\mathcal{I}_{p,q}}(a)].$$

Then it holds in $V^{\mathbb{P}_{\text{max}}}$ that $(H_{\omega_2}, \in, \text{NS}_{\omega_1}, A) \models \forall x, \phi(x)$.

Proof.

- 1° Let g be V -generic for $\mathbb{P}_{\text{max}}$, let us work in $V[g]$, and let $a \in H_{\omega_2}$. We want to show that $(H_{\omega_2}, \in, \text{NS}_{\omega_1}, A) \models \phi[a]$.
- 2° By Proposition 4.8, there exist $p \in g$ and $b \in H_{\omega_2}^p$ such that $a = i^{\mathcal{I}_{p,g}}(b)$.
- 3° By the assumption 6, there exists $q \in g$ such that
 - a. $q \leq p$,
 - b. q is A -iterable,
 - c. $(H_{\omega_2}^q, \in, I^q, A \cap q) \models \phi[i^{\mathcal{I}_{p,q}}(b)]$.

4° It follows by elementarity that

$$(H_{\omega_2}^{\mathcal{P}_{\infty}^{\mathcal{I}_{q,g}}}, \in, i^{\mathcal{I}}(I^q), i^{\mathcal{I}_{q,g}}(A \cap q)) \models \phi[i^{\mathcal{I}_{p,g}}(b)].$$

5° Since q is A -iterable, we conclude that

$$(H_{\omega_2}^{\mathcal{P}_{\infty}^{\mathcal{I}_{q,g}}}, \in, \text{NS}_{\omega_1} \cap \mathcal{P}_{\infty}^{\mathcal{I}_{q,g}}, A \cap \mathcal{P}_{\infty}^{\mathcal{I}_{q,g}}) \models \phi[a].$$

6° The fact that $\phi(x)$ is Σ_1 now implies that

$$(H_{\omega_2}, \in, \text{NS}_{\omega_1}, A) \models \phi[a].$$

□

Theorem 5.2. *Suppose*

1. *there exists a proper class of Woodin cardinals,*

2. A is a universally Baire set of reals,
3. ϕ is a Π_2 sentence in the language $\{R_{\in}^2, R_{\text{NS}_{\omega_1}}^1, R_A^1\}$,
4. there exists a poset that forces “ $(H_{\omega_2}, \in, R_{\text{NS}_{\omega_1}}^1, A_{\dot{g}}) \models \phi$ ”.

Then it holds in $L(\mathbb{R}, A)^{\mathbb{P}_{\max}}$ that $(H_{\omega_2}, \in, \text{NS}_{\omega_1}, A) \models \phi$.

Proof.

- 1° Let $\psi(x)$ be the Σ_1 formula such that $\phi \equiv \forall x, \psi(x)$. By Propositions 2.8, 4.4, and 5.1, it suffices to verify that for all $p \in \mathbb{P}_{\max}$ and all $a \in H_{\omega_2}^p$, there exists an A -iterable $\mathbb{P}_{\max}$ -condition $q \leq p$ such that

$$(H_{\omega_2}^q, \in, I^q, A \cap q) \models \psi[i^{\mathcal{I}^{p,q}}(a)].$$

- 2° Let

- a. $\mathbb{P}$ be a poset satisfying

$$V^{\mathbb{P}} \models ((H_{\omega_2}, \in, \text{NS}_{\omega_1}, A_{\dot{g}}) \models \phi),$$

- b. $\delta > \text{rank}(\mathbb{P})$ be a limit of Woodin cardinals,
- c. $\theta > \delta$ be inaccessible,
- d. $\pi : (N, \in, \bar{A}, \bar{\mathbb{P}}, \bar{\delta}, p) \xrightarrow{\Sigma_\omega} (H_\theta, \in, A, \mathbb{P}, \delta, p)$ be countable, with N transitive,
- e. g_0 be N -generic for $\bar{\mathbb{P}}$.

Note that

$$(H_{\omega_2}^{N[g_0]}, \in, \text{NS}_{\omega_1}^{N[g_0]}, \bar{A}_{g_0}) \models \phi.$$

- 3° By Lemma 3.7, there exists an iteration $\mathcal{I} \in N[g_0]$ of p of length $\omega_1 + 1$ satisfying that $i^{\mathcal{I}}(I^p) = \text{NS}_{\omega_1}^{N[g_0]} \cap \mathcal{P}_\infty^{\mathcal{I}}$. Observe that

$$(H_{\omega_2}^{N[g_0]}, \in, \text{NS}_{\omega_1}^{N[g_0]}, \bar{A}_{g_0}) \models \psi[i^{\mathcal{I}}(a)].$$

- 4° By Lemma 2.1, there exists a $<\bar{\delta}$ -generic filter g_1 over $N[g_0]$ satisfying that

- a. $\text{NS}_{\omega_1}^{N[g_0][g_1]} \cap N[g_0] = \text{NS}_{\omega_1}^{N[g_0]}$,
- b. NS_{ω_1} is saturated in $N[g_0][g_1]$.

- 5° Observe that

- a. $(H_{\omega_2}^{N[g_0][g_1]}, \in, \text{NS}_{\omega_1}^{N[g_0][g_1]}, \bar{A}_{g_0 * g_1}) \models \psi[i^{\mathcal{I}}(a)],$
- b. $\text{NS}_{\omega_1}^{N[g_0][g_1]} \cap \mathcal{P}_\infty^{\mathcal{I}} = i^{\mathcal{I}}(I^p).$

6° By Lemma 2.2, there exists a $<\bar{\delta}$ -generic filter g_2 over $N[g_0][g_1]$ such that $N[g_0][g_1][g_2] \models \text{MA}_{\omega_1}$.

7° By Lemma 2.5, NS_{ω_1} is precipitous in $N[g_0][g_1][g_2]$. Observe also that

- a. $(H_{\omega_2}^{N[g_0][g_1][g_2]}, \in, \text{NS}_{\omega_1}^{N[g_0][g_1][g_2]}, \bar{A}_{g_0 * g_1 * g_2}) \models \psi[i^{\mathcal{I}}(a)],$
- b. $\text{NS}_{\omega_1}^{N[g_0][g_1][g_2]} \cap \mathcal{P}_{\infty}^{\mathcal{I}} = i^{\mathcal{I}}(I^p).$

8° Let $q := (N[g_0][g_1][g_2], \in, \text{NS}_{\omega_1}^{N[g_0][g_1][g_2]}, i^{\mathcal{I}}(a^p))$. By Proposition 3.5 and Proposition 3.10, q is an A -iterable $\mathbb{P}_{\max}$ -condition.

9° Additionally, $q < p$ and $\mathcal{I}_{p,q} = \mathcal{I}$.

10° By Lemma 2.6, $\bar{A}_{g_0 * g_1 * g_2} = A \cap N[g_0][g_1][g_2] = A \cap q$.

11° Thus, 7°a is equivalent to

$$(H_{\omega_2}^q, \in, I^q, A \cap q) \models \psi[i^{\mathcal{I}_{p,q}}(a)],$$

as required in 1°.

□

Corollary 5.3. *Suppose*

1. *there exists a proper class of Woodin cardinals,*
2. *$(*)_{\text{UB}}$ holds,*
3. *A is a universally Baire set,*
4. *ϕ is a Π_2 sentence in the language $\{R_{\in}^2, R_{\text{NS}_{\omega_1}}^1, R_A^1\}$,*
5. *there exists a poset that forces “ $(H_{\omega_2}, \in, \text{NS}_{\omega_1}, A_{\dot{g}}) \models \phi$ ”.*

Then $(H_{\omega_2}, \in, \text{NS}_{\omega_1}, A) \models \phi$.

Proof. This follows immediately from Theorem 5.2.

□

6 $(*)_{\text{UB}}$ implies $\text{UB} - \text{BMM}^{++}$

Lemma 6.1. *Suppose*

1. *there exists a proper class of Woodin cardinals,*
2. *A is a universally Baire set.*

Then A -iterable conditions are dense in $\mathbb{P}_{\max}$.

Proof. This follows from Proposition 3.10. □

Lemma 6.2. *Suppose*

1. *A is a set of reals,*
2. *p is an A -iterable $\mathbb{P}_{\max}$ condition,*
3. *$q \leq p$.*

Then q is an A -iterable. □

Lemma 6.3. *Suppose*

1. *there exists a proper class of Woodin cardinals,*
2. *$(*)_{\text{UB}}$ holds,*
3. *$a \subseteq \omega_1$ satisfies $\omega_1^{L[a]} = \omega_1$,*
4. *A is a universally Baire set,*
5. *g consists of all $p \in \mathbb{P}_{\max}$ for which there is a generic iteration $\mathcal{I}$ such that $i^{\mathcal{I}}(a^p) = a$ and $i^{\mathcal{I}}(I^p) = \text{NS}_{\omega_1} \cap \mathcal{P}_{\infty}^{\mathcal{I}}$.*

Then g is an $L(\mathbb{R}, A)$ -generic filter for $\mathbb{P}_{\max}$ and $\mathcal{P}(\omega_1) \subseteq L(\mathbb{R}, A)[g]$.

Proof. See [Kas25, Lemma 6.2.7]. □

Proposition 6.4. *Suppose*

1. *there exists a proper class of Woodin cardinals,*
2. *$(*)_{\text{UB}}$ holds,*
3. *A is a universally Baire set,*
4. *$\phi(v_a)$ is a Π_1 formula in the language $\{R_{\in}^2, R_{\text{NS}_{\omega_1}}^1, R_A^1\}$,*
5. *$a \subseteq \omega_1$ is such that $\omega_1^{L[a]} = \omega_1$,*
6. *$(H(\omega_2), \in, \text{NS}_{\omega_1}, A) \models \phi[a]$.*

Then there exist $p \in \mathbb{P}_{\max}$ and a generic iteration $\mathcal{I}$ of p such that

- a. p is A -iterable,
- b. $i^{\mathcal{I}}(a^p) = a$,
- c. for all $q \leq p$, $(H(\omega_2)^q, \in, I^q, A \cap q) \models \phi[a^q]$.

Proof.

1° Let g consist of all $p \in \mathbb{P}_{\max}$ for which there is a generic iteration $\mathcal{I}$ satisfying that $i^{\mathcal{I}}(a^p) = a$ and $i^{\mathcal{I}}(I^p) = \mathbf{NS}_{\omega_1} \cap \mathcal{P}_{\infty}^{\mathcal{I}}$. Lemma 6.3 implies that g is an $L(\mathbb{R}, A)$ -generic filter for $\mathbb{P}_{\max}$ and $\mathcal{P}(\omega_1) \subseteq L(\mathbb{R}, A)[g]$.

2° Observe that

$$L(\mathbb{R}, A)[g] \models ((H(\omega_2), \in, \mathbf{NS}_{\omega_1}, A) \models \phi[a]).$$

Thus, there exists $p_0 \in g$ such that

$$p_0 \Vdash_{\mathbb{P}_{\max}}^{L(\mathbb{R}, A)} (H(\omega_2), \in, \mathbf{NS}_{\omega_1}, A) \models \phi[i^{\mathcal{I}_{p_0, \dot{g}}}(a^{p_0})].$$

3° By Lemma 6.1, we may assume, up to strengthening p_0 , that it is A -iterable.

4° Let D consist of all $p \leq p_0$ such that for all $q \leq p$,

$$(H(\omega_2)^q, \in, I^q, A \cap q) \models \phi[a^q].$$

Since by Lemma 6.2, every condition in D is A -iterable, we will be done if we show that D is dense below p_0 .

5° Let us assume otherwise. Then there exists $p_1 \leq p_0$ such that for all $p \leq p_1$, there exists $q \leq p$ such that

$$(H(\omega_2)^q, \in, I^q, A \cap q) \models \neg \phi[a^q]$$

6° For all $q \leq p_0$, if $(H(\omega_2)^q, \in, I^q, A \cap q) \models \neg \phi[a^q]$, then

$$q \Vdash_{\mathbb{P}_{\max}}^{L(\mathbb{R}, A)} ((H(\omega_2), \in, \mathbf{NS}_{\omega_1}, A) \models \neg \phi[i^{\mathcal{I}_{q, \dot{g}}}(a^q)]).$$

(Note that q is A -iterable by Lemma 6.2.)

7° Thus, for all $p \leq p_1$, there exists $q \leq p$ such that

$$q \Vdash_{\mathbb{P}_{\max}}^{L(\mathbb{R}, A)} ((H(\omega_2), \in, \mathbf{NS}_{\omega_1}, A) \models \neg \phi[i^{\mathcal{I}_{q, g}}(a^q)]).$$

8° This means that

$$p_1 \Vdash_{\mathbb{P}_{\max}}^{L(\mathbb{R}, A)} ((H(\omega_2), \in, \mathbf{NS}_{\omega_1}, A) \models \neg \phi[i^{\mathcal{I}_{p_1, g}}(a^{p_1})]),$$

which contradicts the fact that $p_1 \leq p_0$.

□

Theorem 6.5. *Suppose there exists a proper class of Woodin cardinals and $(*)_{\text{UB}}$ holds. Then $\text{UB} - \text{BMM}^{++}$ holds.*

Proof.

- 1° Let A be a universally Baire set, let $\phi(v_a)$ be a Π_1 formula in the language $\{R_{\in}^2, R_{\text{NS}_{\omega_1}}^1, R_A^1\}$, and let $a \in H(\omega_2)$. We want to show that

$$\forall \mathbb{P} \text{ ssp}, V^{\mathbb{P}} \models ((H(\omega_2), \in, \text{NS}_{\omega_1}, A_{\dot{G}}) \models \phi[a]).$$

- 2° We may assume without loss of generality that $a \subseteq \omega_1$ and $\omega_1^{L[a]} = \omega_1$.

- 3° Let $\mathbb{P}$ be an ssp poset and let us assume towards a contradiction that

$$\not\models_{\mathbb{P}} (H(\omega_2), \in, \text{NS}_{\omega_1}, A_{\dot{G}}) \models \phi[a].$$

By restricting the poset, we may assume without loss of generality that

$$\Vdash_{\mathbb{P}} (H(\omega_2), \in, \text{NS}_{\omega_1}, A_{\dot{G}}) \models \neg\phi[a].$$

Furthermore, by doing a two-step iteration if necessary, we may assume that $\mathbb{P}$ forces that NS_{ω_1} is saturated.

- 4° Proposition 6.4 implies that there are $p \in \mathbb{P}_{\max}$ and a generic iteration $\mathcal{I}$ of p such that

- a. p is A -iterable,
- b. $i^{\mathcal{I}}(a^p) = a$,
- c. for all $q \leq p$, $(H(\omega_2)^q, \in, I^q, A \cap q) \models \phi[a^q]$.

- 5° Let G be a V -generic filter for $\mathbb{P}$, let κ be inaccessible in $V[G]$, and let H be $V[G]$ -generic for $\text{Col}(\omega, \kappa)$. We work in $V[G][H]$.

- 6° Let $q_0 := (H_{\kappa}, \in, \text{NS}_{\omega_1}, i^{\mathcal{I}}(a^p))^{V[G]}$. Observe that $q_0 \in \mathbb{P}_{\max}$ and $q_0 < p$ (as witnessed by $\mathcal{I}$).

- 7° By property 4°c and by generic absoluteness, for all $q \leq p$,

$$(H(\omega_2)^q, \in, I^q, A_{G*H} \cap q) \models \phi[a^q].$$

- 8° Since $q_0 < p$, we have that

$$(H(\omega_2)^{q_0}, \in, I^{q_0}, A_{G*H} \cap q_0) \models \phi[a^{q_0}].$$

- 9° Now, note that

- a. $H(\omega_2)^{q_0} = H(\omega_2)^{V[G] \restriction \kappa} = H(\omega_2)^{V[G]}$,

- b. $I^{q_0} = \mathbf{NS}_{\omega_1}^{V[G]},$
- c. $A_{G*H} \cap q_0 = A_{G*H} \cap V[G] = A_G,$
- d. $a^{q_0} = i^{\mathcal{I}}(a^p) = a.$

10° Thus,

$$(H(\omega_2)^{V[G]}, \in, \mathbf{NS}_{\omega_1}^{V[G]}, A_G) \models \phi[a],$$

which contradicts the assumption that

$$\Vdash_{\mathbb{P}}^V (H(\omega_2), \in, \mathbf{NS}_{\omega_1}, A_{\dot{G}}) \models \neg\phi[a].$$

□

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